

1. Project Introduction

The Nifty100 Analytics Platform is a financial analytics system designed to analyze Nifty100 companies using fundamental, valuation, profitability, cashflow and peer comparison metrics. The platform combines ETL pipelines, SQLite storage, FastAPI services and a Streamlit dashboard to provide analyst-friendly insights.

2. System Overview

The system contains four major layers: 1. ETL Layer: Loads and cleans financial datasets. 2. Analytics Engine: Calculates financial ratios including ROE, ROCE, margins, cashflow metrics and CAGR. 3. API Layer: Provides company, screener, sector and peer endpoints. 4. Dashboard Layer: Provides interactive visual analysis.

3. Streamlit Dashboard Navigation

The Streamlit dashboard contains multiple screens: Home: Provides overall platform overview. Company Profile: Displays company fundamentals, ratios and financial information. Screener: Allows filtering companies using financial conditions. Peers: Provides peer comparison analysis. Trends: Shows historical financial trends. Sectors: Provides sector-level analysis. Capital: Displays capital allocation insights. Reports: Generates analyst reports and PDF tearsheets.

4. Using the Screener

The screener helps analysts identify companies based on financial quality parameters. Available filters include: - Minimum ROE - Maximum Debt-to-Equity - Minimum Free Cash Flow - Sector filtering - Revenue CAGR - PAT CAGR - Maximum PE ratio Apply filters and review the ranked company results.

5. Company Profile Analysis

The Company Profile page provides: - Company information - Profitability ratios - ROE and ROCE - Debt metrics - Cashflow metrics - Growth indicators - Historical financial information Select a ticker to view detailed analysis.

6. Sector and Peer Analysis

Sector Analysis provides: - Sector company count - Median valuation metrics - Sector comparisons
Peer Analysis compares companies within similar business groups using financial KPIs.

7. Generating PDF TearSheets

Analysts can generate PDF tearsheets from the Reports section. A tearsheet contains: - Company overview - Financial ratios - Growth metrics - Valuation information - Analyst insights Use the report option from the dashboard to create PDFs.

8. API Usage Guide

The FastAPI service provides programmatic access. Base URL: <http://127.0.0.1:8000> Important endpoints: [/api/v1/health](#) [/api/v1/companies](#) [/api/v1/screener](#) [/api/v1/sectors](#) [/api/v1/peers](#)

9. API Curl Examples

Health Check: curl http://127.0.0.1:8000/api/v1/health Company Data: curl
http://127.0.0.1:8000/api/v1/companies/TCS Screener: curl
"http://127.0.0.1:8000/api/v1/screener?min_roe=15"

10. Troubleshooting Guide

Common issues: API not starting: Check if port 8000 is already occupied. Dashboard not loading: Ensure Streamlit is running on port 8501. Database errors: Verify db/nifty100.db exists. Missing packages: Activate the virtual environment and install requirements. For API issues check /docs Swagger documentation.